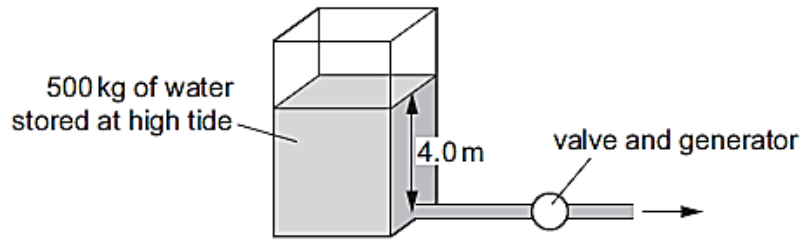


8 A simple idea for generating electricity from the tides allows water stored in a container at high tide to flow away through a generator at low tide.

7



At high tide 500 kg of water is stored to a height of 4.0 m. When the valve is opened the container empties in 5.0 s and the generator operates with an efficiency of 40%.

What is the average electrical power generated during the period of water flow?

- A** 780 W **B** 1600 W **C** 2000 W **D** 3900 W

Ans: A

$$P_{out} = \frac{mgh}{t} \times 0.40$$

$$P_{out} = \frac{500(9.81)(4.0 / 2)}{5.0} \times 0.40 = 784.8 = 780 \text{ W}$$